

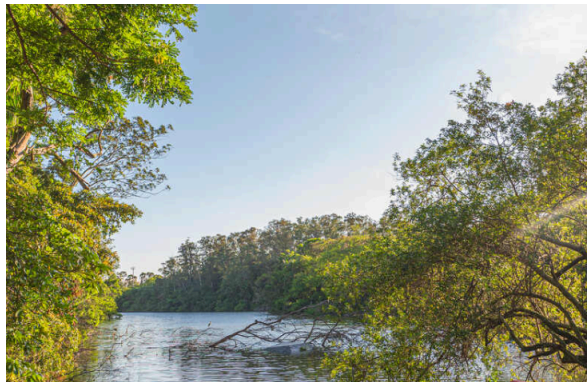
Pathways of Elements in Ecosystems

Chapter Concepts

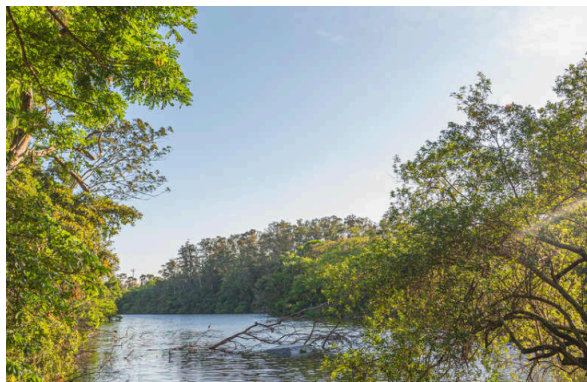
- **Energy transformations** and **element cycling** are intimately linked.
- **Ecosystems** can be modeled as a series of linked **compartments**.
- **Water** provides a physical model of element cycling in ecosystems.
- The **carbon cycle** is closely tied to the flux of **energy** through the biosphere.
- **Nitrogen** assumes many **oxidation states** in its cycling through ecosystems.
- The **phosphorus cycle** is chemically uncomplicated.
- **Sulfur** exists in many **oxidized** and **reduced** forms.
- **Microorganisms** assume diverse roles in element cycles.

Energy Transformations and Element Cycling

- **Organisms** move chemical elements through their cycles by carrying out **biochemical transformations**.
- **Assimilatory processes** incorporate inorganic elements into organic molecules (e.g., **photosynthesis** converts CO_2 to carbohydrates).
- **Dissimilatory processes** (e.g., **respiration**) return elements to inorganic forms, releasing energy.
- Most **biochemical energy transformations** involve **oxidation** and **reduction** of **carbon**, **oxygen**, **nitrogen**, and **sulfur**.
- **Energy-releasing oxidations** are paired with **energy-requiring reductions**; energy shifts from reactants to products.



- Coupled transformations are the basis of energy flow in ecosystems.
- Biochemical transformations are linked into **biochemical pathways**.



Ecosystem Compartment Models

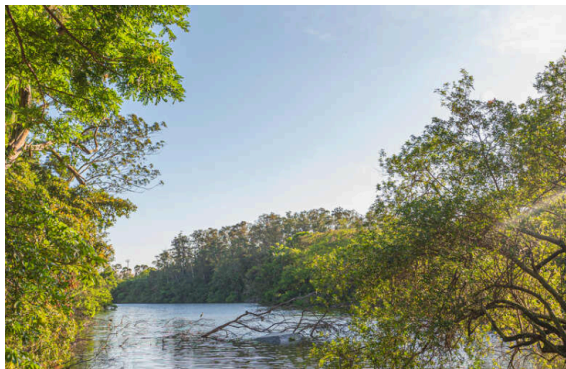
- Each form of an element resides in a separate **compartment** (e.g., inorganic, organic, geologic).
- **Photosynthesis** moves carbon from inorganic to organic compartments; **respiration** returns it.
- Compartment models can be hierarchical, with subcompartments (e.g., autotrophs, animals, detritus, microbes).
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- Elements cycle rapidly among some compartments, slowly among others (e.g., coal, oil, limestone are slow geologic compartments).

The Hydrologic Cycle as a Model

- **Water** cycles through the biosphere via **evaporation**, **transpiration**, **precipitation**, and **runoff**.
- Solar energy drives evaporation; condensation releases heat.
- The **hydrologic cycle** illustrates basic features of element cycles.
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- Water in the atmosphere has a short residence time; in oceans and lakes, much longer.

The Carbon Cycle

- The **carbon cycle** is driven by solar energy and involves complex chemical reactions.
- Three main processes: (1) **photosynthesis** and **respiration**, (2) **exchange** between atmosphere and oceans, (3) **precipitation** of carbonate sediments.

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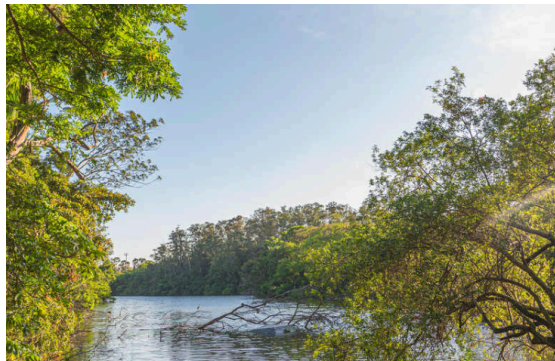
- Carbon cycles between **organic** and **inorganic** forms.

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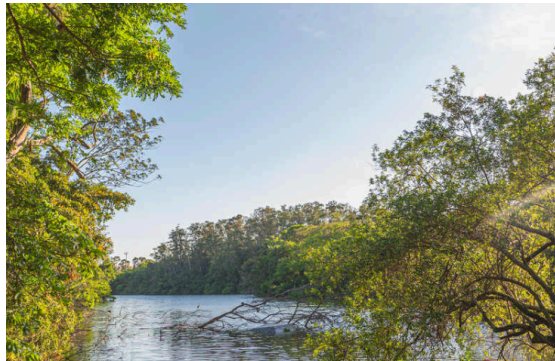
- Most carbon is locked in **sedimentary rocks**.

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- Biological and geologic processes regulate atmospheric CO_2 over time.

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- Atmospheric CO_2 has declined since the Paleozoic era.

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- Fossil soils and plant evolution reveal changes in atmospheric CO_2 .

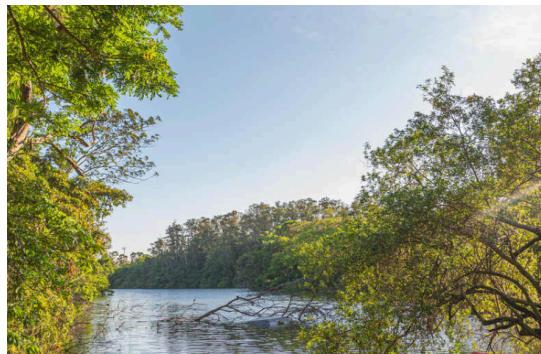
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The Nitrogen Cycle

- **Nitrogen** cycles through many **oxidation states** and follows complex pathways.
- The largest pool is atmospheric N_2 .
- **Nitrogen fixation** by bacteria converts N_2 to usable forms.

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- Plants assimilate nitrogen as **ammonium** (NH_4^+) or **nitrate** (NO_3^-).
- **Ammonification** converts organic nitrogen to ammonia.
- **Nitrification** oxidizes ammonia to nitrite (NO_2^-) and nitrate.
- **Denitrification** returns nitrogen to the atmosphere as N_2 .
- **Rhizobium** bacteria in root nodules fix nitrogen in symbiosis with plants.

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- Stable isotopes (e.g., ^{15}N) are used to trace nitrogen movement in ecosystems.

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The Phosphorus Cycle

- **Phosphorus** is required for nucleic acids, membranes, energy transfer, bones, and teeth.
- The phosphorus cycle is simpler than nitrogen; phosphorus does not undergo redox reactions.
- Plants assimilate phosphorus as **phosphate** (PO_4^{3-}).

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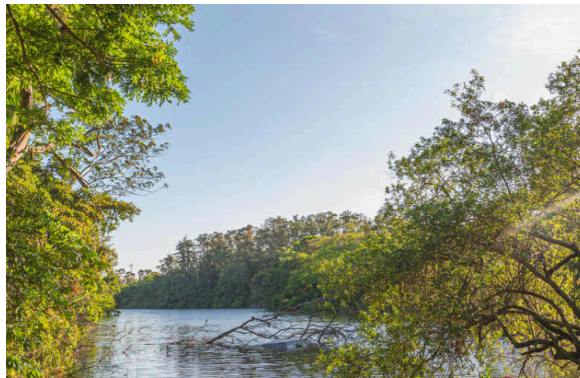
- Phosphorus often limits primary production in aquatic systems.
- Cycling differs from carbon: phosphorus is removed from water column as insoluble compounds.

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- Carbon:phosphorus (C:P) ratios reflect phosphorus limitation in food webs.

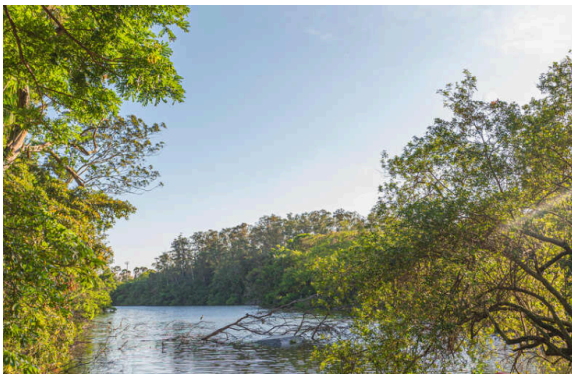
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The Sulfur Cycle

- **Sulfur** is required for amino acids and cycles through many oxidized and reduced forms.
- The most oxidized form is **sulfate** (SO_4^{2-}); the most reduced are **hydrogen sulfide** (H_2S) and organic sulfur.

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- Under anaerobic conditions, sulfate can act as an oxidizer; reduced sulfur forms can accumulate.
- Sulfides in coal and oil, when exposed to air, oxidize to sulfate and form sulfuric acid.

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Microorganisms in Element Cycles

- **Microorganisms** (especially bacteria and archaea) drive many transformations in element cycles.
- Specialized bacteria can metabolize substrates under anaerobic conditions and use inorganic compounds as energy sources.
- Chemoautotrophic sulfur bacteria form the base of food chains in hydrothermal vent communities.

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